

PDF → Lean 4 → L^AT_EX Round-Trip Demo

May 10, 2026

1 Introduction

This document was produced by typesetting a `.tex` source with `pdflatex`, then feeding the resulting PDF (after `mutool clean -d`) into our Lean-4 parser, which emitted a new `.tex` that was again compiled with `pdflatex`. If you can read this paragraph, the round trip succeeded.

2 What the parser handles

- Lexical layer: numbers, names, literal strings, hex strings, comments.
- All eight PDF object types, plus indirect references.
- Cross-reference *tables* and (uncompressed) cross-reference *streams*; `Prev` chains for incremental updates.
- Object-stream resolution (`/Type /ObjStm`).
- Page tree walk and content-stream parsing.
- Text-state operators: `BT`, `ET`, `Tf`, `Td`, `TD`, `Tm`, `T*`, `Tj`, `TJ`, `'`, `"`.
- Graphics-state nesting (`q/Q`) and `cm`.

3 Out of scope (preprocess required)

- Flate/LZW decompression — run `mutool clean -d` first.
- Encryption (RC4 / AES).
- Font CMaps — only the predefined ASCII encodings render correctly.
- Vector graphics, images, colour.

4 Page Two

A few mathematical expressions to push the parser:

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \qquad \sum_{k=0}^n nk = 2^n.$$

A bullet list:

- alpha
- beta
- gamma

A numbered list:

1. one
2. two
3. three